

COMP0078 Exam Revision Guide

Proof Skeletons, Key Concepts & Likely Exam Variations

Based on Mark Herbster's advice: "I would tend to ask a question involving applying your insight to prove a similar result in a different situation"

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Priority Key: [MUST] = Must know cold (will lose significant marks without it). [SHOULD] = Should know (likely tested, possibly in a variant form). [NICE] = Nice to know (context, unlikely to be tested directly).

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1 Lecture 1: Introduction

Topic	Priority	Skeleton / Salient Idea	Likely Exam Form
Normal equations	[SHOULD]	$\nabla_{\mathbf{w}} \ X\mathbf{w} - \mathbf{y}\ ^2 = 0 \Rightarrow \hat{\mathbf{w}} = (X^\top X)^{-1} X^\top \mathbf{y}$	Derive for a modified objective (e.g. weighted LS)
Bayes estimator (squared loss)	[MUST]	Differentiate $\sum_y (y - z)^2 p(y x)$ w.r.t. z ; get $f^*(x) = \mathbb{E}[y x]$	“Find Bayes estimator for L_1 loss” $\rightarrow$ median
Bias-variance decomposition	[MUST]	Use $\mathbb{E}[Z^2] = \text{Var}(Z) + \mathbb{E}[Z]^2$ on both y' and $A(\mathbf{x}')$; independence kills cross term; regroup into 3 terms	State decomposition, explain each term, or derive for a variant loss
1-NN asymptotic bound	[SHOULD]	$\mathcal{E}_{1\text{-NN}} \leq 2\mathcal{E}^*$; equality cases: $p^* = 1$ or $p^* = 1/C$	Give bound and example achieving equality
k-NN: $k(m) \rightarrow \infty$, $k/m \rightarrow 0$	[SHOULD]	Conditions for asymptotic optimality	True/False: “larger k overfits” (False)
NFL Theorem	[NICE]	Statement only (proof not examined)	Explain why prior knowledge (H) is needed
Cross-validation	[SHOULD]	K -fold procedure; LOO = $K = m$	When to use; how to select hyperparameters

Bias-Variance Proof Skeleton:

1. Expand $\mathbb{E}[(y' - A(\mathbf{x}'))^2]$ using independence of y' and $A(\mathbf{x}')$.
2. Apply $\mathbb{E}[Z^2] = \text{Var}(Z) + \mathbb{E}[Z]^2$ to both y' and $A(\mathbf{x}')$.
3. Middle cross term: $-2\mathbb{E}[y']\mathbb{E}[A]$
4. Regroup: $\text{Var}(y') + (f^* - \mathbb{E}[A])^2 + \text{Var}(A)$.

$$-2\mathbb{E}[y']\mathbb{E}[A] =$$

□

Key insight: Only works for squared loss. Caveat is examinable.

2 Lecture 2: Kernels & Ridge Regression

Topic	Priority	Skeleton / Salient Idea	Likely Exam Form
Ridge primal solution	[MUST]	$\nabla = 0 \Rightarrow (X^\top X + \lambda I_n)\mathbf{w} = X^\top \mathbf{y}$	Derive for weighted/Tikhonov variant
Ridge dual solution	[MUST]	Stationarity $\Rightarrow \mathbf{w} = X^\top \boldsymbol{\alpha} / \lambda$, substitute back $\Rightarrow (XX^\top + \lambda I_m)\boldsymbol{\alpha} = \mathbf{y}$	Derive dual for modified regularizer; kernelized version
Primal vs dual complexity	[SHOULD]	Primal: $O(n^3)$; Dual: $O(m^3)$. Use dual when $n \gg m$	“When would you use dual?”
Kernel $\Leftrightarrow$ PSD ($\Leftarrow$ direction)	[MUST]	$\sum_{i,j} c_i c_j K(x_i, x_j) = \ \sum_i c_i \phi(x_i)\ ^2 \geq 0$	Prove a specific K is PSD by exhibiting ϕ
Schur product theorem	[SHOULD]	$A = UU^\top, B = VV^\top$; $\sum_{i,j} z_i z_j A_{ij} B_{ij} = \sum_{r,s} (\sum_i z_i U_{ir} V_{is})^2 \geq 0$	“Prove $K_1 K_2$ is a kernel”
Kernel closure properties	[MUST]	Sum, scaling, product, composition, $f(x)K(x,t)f(t)$	“Is $K(x,t) = \dots$ a valid kernel?” Give ϕ or counterexample
Polynomial kernel $\rightarrow$ feature map	[MUST]	Expand $(1 + x^\top t)^2$; cross-term coeff $2 \rightarrow \sqrt{2}$ in each ϕ component	Given K , find explicit ϕ
Gaussian kernel	[SHOULD]	PSD (Taylor series of exp); feature map is ∞ -dim	“Why can’t Gaussian have finite-dim ϕ ?”

Ridge Dual Proof Skeleton:

1. Stationarity: $\mathbf{w} = \frac{1}{\lambda} X^\top (\mathbf{y} - X\mathbf{w}) = X^\top \boldsymbol{\alpha}$ where $\alpha_i := (y_i - \mathbf{w}^\top \mathbf{x}_i) / \lambda$.
2. Substitute $\mathbf{w} = X^\top \boldsymbol{\alpha}$ back into α_i definition.
3. Get $(XX^\top + \lambda I_m)\boldsymbol{\alpha} = \mathbf{y}$.
4. Prediction: $f(\mathbf{x}) = \sum_i \alpha_i (\mathbf{x}_i^\top \mathbf{x})$ — only inner products!

□

Key insight: Optimal $\mathbf{w}$ lives in $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ (Representer Theorem preview).

3 Lecture 3: Support Vector Machines

Topic	Priority	Skeleton / Salient Idea	Likely Exam Form
Canonical parameterization	[SHOULD]	$\min_i y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1 \Rightarrow \text{margin} = 1/\ \mathbf{w}\ $	Explain normalized vs canonical
Hard margin primal $\rightarrow$ dual	[MUST]	$L = \frac{1}{2}\ \mathbf{w}\ ^2 - \sum \alpha_i [y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1]$. Stationarity: $\mathbf{w} = \sum \alpha_i y_i \mathbf{x}_i$, $\sum \alpha_i y_i = 0$. Sub back: dual $Q = \sum \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j$	Derive dual for a variant (e.g. different regularizer, kernel version)
KKT complementary slackness (hard)	[MUST]	$\alpha_i [y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1] = 0$. $\alpha_i > 0 \Rightarrow$ on margin	“Which points are SVs?”
Soft margin: box constraint	[MUST]	$\partial L / \partial \xi_i = 0 \Rightarrow C - \alpha_i - \beta_i = 0 \Rightarrow 0 \leq \alpha_i \leq C$	KKT trichotomy: $\alpha_i = 0$ vs $(0, C)$ vs C
SVM $\equiv$ hinge + L2 reg	[MUST]	Optimal $\xi_i^* = \max(0, 1 - y_i \mathbf{w}^\top \mathbf{x}_i)$; $\lambda = 1/(2C)$	Given a new loss+reg combo, identify equivalence
Kernel SVM	[SHOULD]	Replace $\mathbf{x}_i^\top \mathbf{x}_j \rightarrow K(\mathbf{x}_i, \mathbf{x}_j)$ everywhere; sparsity from SVs	Write kernelized decision function
SV generalization bound	[NICE]	$\mathbb{E}[\text{error}] \leq n_{\text{sv}}/m$	State and interpret

Hard Margin Dual Proof Skeleton:

1. Write Lagrangian with $\alpha_i \geq 0$.
2. $\partial L / \partial b = 0 \Rightarrow \sum \alpha_i y_i = 0$. $\partial L / \partial \mathbf{w} = 0 \Rightarrow \mathbf{w} = \sum \alpha_i y_i \mathbf{x}_i$.
3. Substitute back: $\frac{1}{2} \mathbf{w}^\top \mathbf{w} = \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j$; b term vanishes; constant $\sum \alpha_i$ remains.
4. Result: $\max_{\alpha} \sum \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j$ s.t. $\alpha_i \geq 0$, $\sum \alpha_i y_i = 0$. □

Key insight: Only inner products appear $\rightarrow$ kernel trick applicable.

4 Lecture 4: Trees & Boosting

Topic	Priority	Skeleton / Salient Idea	Likely Exam Form
Greedy recursive partitioning	[MUST]	Enumerate (j, s) ; each region predicts mean; pick min total squared error	Hand-compute a regression tree (very likely)
Impurity measures	[SHOULD]	Misclassification, Gini, cross-entropy. Gini/entropy more sensitive than misclass	“Why prefer Gini over misclass for growing?”
Bagging vs Boosting	[MUST]	Bagging: equal weight, parallel, variance reduction. Boosting: adaptive weight, sequential, bias reduction	Compare and contrast
Random Forest	[SHOULD]	Bagging + random feature subset ($\sqrt{n}$ or $\log n$) at each split $\rightarrow$ decorrelate trees	Explain the two decorrelation mechanisms
AdaBoost algorithm	[MUST]	$D_1 = 1/m$; $\epsilon_t = \sum D_t(i) \mathbf{1}[h_t \neq y_i]$; $\alpha_t = \frac{1}{2} \ln \frac{1-\epsilon_t}{\epsilon_t}$; $D_{t+1} \propto D_t \exp(-\alpha_t y_i h_t)$	Write algorithm; explain weight update
AdaBoost training error bound	[MUST]	$\leq \prod_t Z_t = \prod_t 2\sqrt{\epsilon_t(1-\epsilon_t)} \leq e^{-2\gamma^2 T}$	Prove $\prod Z_t$ bound or a variant
Optimal α_t derivation	[SHOULD]	$dZ_t/d\alpha_t = 0 \Rightarrow \alpha_t = \frac{1}{2} \ln \frac{1-\epsilon_t}{\epsilon_t}$	Derive α_t for a modified Z_t
Boosting = exp loss minimizer	[SHOULD]	Forward stagewise: $D_t(i) \propto e^{-y_i f^{(t-1)}(\mathbf{x}_i)}$	Explain connection
Weak learner / weakly-universal	[SHOULD]	$\epsilon_t < 1/2$ for any distribution. Decision stumps: NOT w.u. (XOR counterexample)	Prove/disprove for a given $\mathcal{H}$

AdaBoost Training Error Proof Skeleton:

- $\mathbf{1}[H \neq y_i] \leq e^{-y_i f(\mathbf{x}_i)}$ (0-1 $\leq$ exponential).
- Telescope $D_{T+1}(i)$: $D_{T+1}(i) = e^{-y_i f(\mathbf{x}_i)} / (m \prod Z_t)$.
- Since $\sum D_{T+1}(i) = 1$: $\frac{1}{m} \sum e^{-y_i f} = \prod Z_t$.
- Compute Z_t : split into correct/incorrect $\rightarrow Z_t = \epsilon_t e^{\alpha_t} + (1 - \epsilon_t) e^{-\alpha_t} = 2\sqrt{\epsilon_t(1 - \epsilon_t)}$.
- Use $\sqrt{1 - 4\gamma_t^2} \leq e^{-2\gamma_t^2}$ (from $1 - x \leq e^{-x}$). □

Key insight: $Z_t < 1$ iff weak learner beats random $\rightarrow$ exponential convergence.

5 Lecture 5: Online Learning I

Topic	Priority	Skeleton / Salient Idea	Likely Exam Form
Online model / regret	[MUST]	Adversarial, no distributional assumptions; regret = $L_A - \min_i L_i$	Define; contrast with batch
Halving	[SHOULD]	Majority of consistent experts; $M \leq \log_2 n$ (halve survivors each mistake)	State bound and one-line proof
WM bound	[SHOULD]	Upper: $W_{t+1} \leq \frac{1+\beta}{2} W_t$ per mistake. Lower: $\beta^{M_i} \leq W_{m+1}$. Squeeze.	Prove for a modified β schedule
WAA theorem (entropic loss)	[MUST]	Exact equality: $L_{en}(y_t, \hat{y}_t) - \sum u_i L_{en}(y_t, x_{t,i}) = d(\mathbf{u}, \mathbf{v}_t) - d(\mathbf{u}, \mathbf{v}_{t+1})$. Telescope $\rightarrow \ln n$.	Apply WAA to new loss; derive bound for square loss ($\eta = 1/2$)
Hedge theorem	[MUST]	KL potential; $e^{-x} \leq 1 - x + x^2/2$; $\ln(1+x) \leq x$; telescope; optimize $\eta = \sqrt{2 \ln n / m} \rightarrow \sqrt{2m \ln n}$	“Apply Hedge to a variant setting” (Mark’s stated style)
Perceptron bound (Novikoff)	[MUST]	Upper: $\ \mathbf{w}_{t+1}\ ^2 \leq \ \mathbf{w}_t\ ^2 + R^2$ (cross ≤ 0). Lower: $\ \mathbf{w}_t\ \geq M\gamma$ (C-S). Squeeze: $M \leq R^2/\gamma^2$.	Modified update rule or kernel version
OGD regret bound	[MUST]	Expand $\ \mathbf{w}_{t+1} - \mathbf{u}\ ^2$; rearrange; telescope; optimize $\eta = U/(R\sqrt{m}) \rightarrow UR\sqrt{m}$	Derive for different convex loss
Winnow	[SHOULD]	Multiplicative update gated by $x_{t,i}$; $M \leq 3k(\log n + 1) + 2$. Key: $O(k \log n)$ vs Perceptron $O(kn)$	Compare to Perceptron for sparse hypothesis
Boolean / DNF	[SHOULD]	Literal, conjunction, disjunction, DNF; Anova kernel for DNF	Definitions + bounds
Online-to-batch	[NICE]	Average online iterates; bound via online regret	State conversion
Fixed Share	[NICE]	Tracking best expert; shifting loss bound; proof sketch via meta-experts	Unlikely full proof; concept only

Hedge Proof Skeleton (template for all variants):

1. **Potential:** $\Phi_t = d(\mathbf{u}, \mathbf{v}_t) = \sum u_i \ln(u_i/v_{t,i})$ (KL divergence).
2. **Progress:** $\Phi_t - \Phi_{t+1} = -\eta \mathbf{u} \cdot \boldsymbol{\ell}_t - \ln Z_t$.
3. **Linearize** Z_t : $e^{-x} \leq 1 - x + x^2/2$ then $\ln(1+x) \leq x$.
4. **Result:** $\Phi_t - \Phi_{t+1} \geq \eta(\mathbf{v}_t \cdot \boldsymbol{\ell}_t - \mathbf{u} \cdot \boldsymbol{\ell}_t) - \frac{\eta^2}{2} \sum v_{t,i} \ell_{t,i}^2$.
5. **Telescope:** Regret $\leq \frac{\Phi_1}{\eta} + \frac{\eta}{2} \sum_t \sum_i v_{t,i} \ell_{t,i}^2$.
6. **Bound:** $\Phi_1 \leq \ln n$, $\ell_{t,i}^2 \leq 1 \Rightarrow \leq \frac{\ln n}{\eta} + \frac{\eta m}{2}$.
7. **Optimize:** $\eta^* = \sqrt{2 \ln n / m} \Rightarrow \text{Regret} \leq \sqrt{2m \ln n}$. □

To adapt: Change step 6 for different loss range $[0, B] \rightarrow B^2$ factor. For bandit, $\ell_{t,i}^2$ bound changes to n (see L8).

Perceptron Proof Skeleton:

1. **Upper:** On mistake, $\|\mathbf{w}_{t+1}\|^2 = \|\mathbf{w}_t\|^2 + 2y_t(\mathbf{w}_t \cdot \mathbf{x}_t) + \|\mathbf{x}_t\|^2 \leq \|\mathbf{w}_t\|^2 + R^2$ (cross ≤ 0 on mistake). Induction: $\leq MR^2$.
2. **Lower:** $\mathbf{w}_{t+1} \cdot \mathbf{v} \geq \mathbf{w}_t \cdot \mathbf{v} + \gamma$. Induction: $\geq M\gamma$. By C-S: $\|\mathbf{w}\| \geq M\gamma$.
3. **Squeeze:** $(M\gamma)^2 \leq MR^2 \Rightarrow M \leq R^2/\gamma^2$. □

To adapt: Change update rule $\rightarrow$ recompute upper bound. Change margin $\rightarrow$ recompute lower bound.

OGD Proof Skeleton:

1. **Convexity:** $L(y, \hat{y}) - L(y, \mathbf{u} \cdot \mathbf{x}) \leq (\mathbf{w}_t - \mathbf{u}) \cdot \mathbf{z}_t$ (subgradient).
2. **Expand:** $\|\mathbf{w}_{t+1} - \mathbf{u}\|^2 = \|\mathbf{w}_t - \mathbf{u}\|^2 - 2\eta(\mathbf{w}_t - \mathbf{u}) \cdot \mathbf{z}_t + \eta^2 \|\mathbf{z}_t\|^2$.
3. **Rearrange + telescope:** $\sum (\mathbf{w}_t - \mathbf{u}) \cdot \mathbf{z}_t \leq \frac{U^2}{2\eta} + \frac{\eta m R^2}{2}$.
4. **Optimize:** $\eta = U/(R\sqrt{m}) \Rightarrow UR\sqrt{m}$. □

Parallel to Hedge: potential $\|\cdot\|^2$ replaces KL; expand-square replaces e^{-x} trick.

6 Lecture 6: Graph-based Semi-supervised Learning

Topic	Priority	Skeleton / Salient Idea	Likely Exam Form
SSL vs SL vs UL	[SHOULD]	SL: labeled only. UL: unlabeled only. SSL: both.	Define and compare
Graph Laplacian $L = D - W$	[MUST]	Definition + three properties (PSD, $L\mathbf{1} = \mathbf{0}$, connected $\Rightarrow$ unique null eigvec)	Compute L for a given graph
Quadratic form proof	[MUST]	$\mathbf{u}^\top L \mathbf{u} = \sum_i d_i u_i^2 - \sum_{i,j} w_{ij} u_i u_j$. Use symmetry to write as $\frac{1}{2} \sum_{i,j} w_{ij} (u_i - u_j)^2$.	Prove from scratch
Spectral clustering	[SHOULD]	cut = $\frac{1}{4} \mathbf{u}^\top L \mathbf{u}$; RatioCut $\rightarrow$ NP-hard $\rightarrow$ relax $\rightarrow$ Fiedler vector	Explain relaxation procedure
Harmonic solution	[MUST]	$\bar{u}_i = \frac{\sum_j w_{ij} \bar{u}_j}{d_i}$ (weighted average of neighbors). Consensus algorithm.	Compute harmonic labels on a given graph
Effective resistance	[SHOULD]	$r_G(p, q) = L_{pp}^+ + L_{qq}^+ - 2L_{pq}^+$ (squared distance in L^+ kernel space)	Use optimization lemma to derive
Random walk interpretation	[SHOULD]	$p_i = \Pr[\text{reach } V_1 \text{ first}] = \text{harmonic solution}$	Explain connection
Two-clique example	[SHOULD]	Dense clique $\rightarrow$ small internal resistance $\rightarrow$ labeling respects clusters	Draw and explain
Kirchhoff Circuit Laws	[NICE]	Not examined	—
Sections VIII–X	[NICE]	Not examined	—

Quadratic Form Proof:

1. $\mathbf{u}^\top L \mathbf{u} = \mathbf{u}^\top (D - W) \mathbf{u} = \sum_i d_i u_i^2 - \sum_{i,j} w_{ij} u_i u_j$.
2. By symmetry of W : $\sum_i d_i u_i^2 = \sum_i (\sum_j w_{ij}) u_i^2 = \frac{1}{2} \sum_{i,j} w_{ij} u_i^2 + \frac{1}{2} \sum_{i,j} w_{ij} u_j^2$.
3. Combine: $= \frac{1}{2} \sum_{i,j} w_{ij} (u_i^2 + u_j^2 - 2u_i u_j) = \frac{1}{2} \sum_{i,j} w_{ij} (u_i - u_j)^2$. □

7 Lecture 7: Learning Theory

Topic	Priority	Skeleton / Salient Idea	Likely Exam Form
Expected vs empirical error	[MUST]	$L_{\mathcal{D}}(h) = \Pr[h(x) \neq y]$; $L_S(h) = \frac{1}{m} \sum \mathbf{1}[h(x_i) \neq y_i]$	Definitions
Validation set bound	[MUST]	Hoeffding on $Z_i = \mathbf{1}[h(x_i) \neq y_i]$; set $\delta = e^{-2m\epsilon^2}$	Derive; explain limitations
PAC model (realizable)	[MUST]	Realizability; ϵ , δ ; sample complexity $m_H(\epsilon, \delta)$	State definition precisely
Finite class PAC bound	[MUST]	$H_B \rightarrow$ misleading $\rightarrow$ union bound $\rightarrow (1 - \epsilon)^m \leq e^{-\epsilon m} \rightarrow m \geq \frac{\log(H /\delta)}{\epsilon}$	Prove or adapt to new setting
VC dimension definition	[MUST]	Shattering; VCdim = $\max\{ C : H \text{ shatters } C\}$	Compute for given H
VC dim examples	[MUST]	Thresholds: 1. Intervals: 2. Rectangles: 4. Halfspaces $\mathbb{R}^n$: n . Finite H : $\leq \log_2 H $.	“Compute VCdim of H ” (shatter + cannot shatter)
Large margin VC dim	[MUST]	VCdim $\leq \Lambda^2 R^2$; larger margin $\rightarrow$ smaller VC $\rightarrow$ better generalization	Connection to SVM; state and interpret
Sauer’s Lemma	[SHOULD]	$\Pi_H(m) \leq (em/d)^d$ for $m > d$. Exponential $\rightarrow$ polynomial transition.	Statement only
Fundamental Theorem	[SHOULD]	$m_H \sim \Theta(d/\epsilon)$. H PAC learnable $\Leftrightarrow$ VCdim $< \infty$. ERM is near-optimal.	Statement only; proof not examined
ϵ -representative sample	[MUST]	$\forall h \in H: L_S(h) - L_{\mathcal{D}}(h) \leq \epsilon$	State definition
Representative $\Rightarrow$ ERM works	[MUST]	Three-step chain: $L_{\mathcal{D}}(h_S) \leq L_S(h_S) + \epsilon/2 \leq L_S(h) + \epsilon/2 \leq L_{\mathcal{D}}(h) + \epsilon$	Prove this lemma (very likely)
Uniform convergence	[MUST]	UC property $\Rightarrow$ agnostic PAC learnable with $m_H(\epsilon, \delta) \leq m_H^{UC}(\epsilon/2, \delta)$	State and apply
Agnostic PAC	[MUST]	$L_{\mathcal{D}}(h_S) \leq \min_H L_{\mathcal{D}}(h) + \epsilon$. Union + two-sided Hoeffding $\rightarrow m \geq \frac{2 \log(2 H /\delta)}{\epsilon^2}$	Prove; contrast rate $O(1/\sqrt{m})$ vs realizable $O(1/m)$
Error decomposition	[MUST]	$\epsilon_{\text{app}} = \min_H L_{\mathcal{D}}(h)$ (shrinks with $ H $); $\epsilon_{\text{est}} = L_{\mathcal{D}}(h_S) - \epsilon_{\text{app}}$ (shrinks with m , grows with $ H $)	State, explain tradeoff

Finite Class PAC Proof Skeleton (Realizable):

1. Define $H_B = \{h : L_{\mathcal{D}}(h) > \epsilon\}$, $M = \{S : \exists h \in H_B, L_S(h) = 0\}$.
2. ERM fails $\Rightarrow S \in M$ (picked a bad-but-consistent h).
3. Union bound: $\Pr[S \in M] \leq \sum_{h \in H_B} \Pr[L_S(h) = 0]$.
4. Each term: $\Pr[L_S(h) = 0] = (1 - L_{\mathcal{D}}(h))^m < (1 - \epsilon)^m \leq e^{-\epsilon m}$.
5. Combine: $< |H|e^{-\epsilon m} \leq \delta$ when $m \geq \log(|H|/\delta)/\epsilon$. □

Agnostic Finite Class Proof Skeleton:

1. Goal: show UC, i.e. $\Pr[\exists h : |L_S - L_{\mathcal{D}}| > \epsilon] < \delta$.

2. Union bound: $\leq \sum_{h \in H} \Pr[|L_S(h) - L_{\mathcal{D}}(h)| > \epsilon]$.
3. Two-sided Hoeffding: each $\leq 2e^{-2m\epsilon^2}$.
4. Combine: $\leq 2|H|e^{-2m\epsilon^2} \leq \delta$ when $m \geq \frac{\log(2|H|/\delta)}{2\epsilon^2}$.
5. Plug into UC $\rightarrow$ agnostic PAC: replace $\epsilon \rightarrow \epsilon/2$, get $m \geq \frac{2\log(2|H|/\delta)}{\epsilon^2}$.

□

Key difference from realizable: Two-sided Hoeffding (not one-sided); rate $1/\epsilon^2$ (not $1/\epsilon$).

Representative Sample Lemma (Three-Step Chain): If S is $\epsilon/2$ -representative:

$$L_{\mathcal{D}}(h_S) \underbrace{\leq}_{\text{rep}} L_S(h_S) + \frac{\epsilon}{2} \underbrace{\leq}_{\text{ERM}} L_S(h) + \frac{\epsilon}{2} \underbrace{\leq}_{\text{rep}} L_{\mathcal{D}}(h) + \epsilon \quad \forall h \in H. \quad \blacksquare$$

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8 Lecture 8: Online Learning II (Bandits / EXP3)

Topic	Priority	Skeleton / Salient Idea	Likely Exam Form
Full vs partial feedback	[MUST]	Full: see all ℓ_t . Partial: see only $\ell_{t,\hat{y}_t}$.	Define both protocols
Exploration vs exploitation	[SHOULD]	Explore: try arms to learn. Exploit: use best known. Tension between them.	Explain trade-off
Unbiased estimator	[MUST]	$\mathbb{E}[\hat{\theta}] = \theta$. German tank example.	Check if a given estimator is unbiased
Importance weighting	[MUST]	$\hat{\theta}_p = X/p$ with prob p , else 0. $\mathbb{E} = X$. Variance cost: divided by p .	Design IW estimator for new setting
Hallucinated loss	[MUST]	$\ell_{t,i}^h = \frac{\ell_{t,i}}{v_{t,i}} \mathbf{1}[i = \hat{y}_t]$. Unbiased: $\mathbb{E}[\ell_{t,i}^h] = \ell_{t,i}$.	Verify unbiasedness; compute second moment
Oblivious adversary	[MUST]	Losses fixed before game starts; needed so $\ell_t \perp \hat{y}_t$ for taking $\mathbb{E}$	Explain why needed
EXP3 = Hedge + hallucinated	[MUST]	Same exponential weight update, but fed ℓ_t^h instead of ℓ_t	State connection
EXP3 theorem & proof	[MUST]	Hedge bound on $\ell_t^h \rightarrow$ take $\mathbb{E} \rightarrow$ three identities $\rightarrow$ variance $\leq n \rightarrow$ optimize $\eta \rightarrow \sqrt{2mn \ln n}$	Apply to variant (e.g. observe k arms, different loss range)
Hedge vs EXP3 comparison	[MUST]	$\sqrt{2m \ln n}$ vs $\sqrt{2mn \ln n}$. Extra $\sqrt{n}$ = price of partial info. Source: $\sum v_{t,i} \ell_{t,i}^h \leq 1$ vs $\mathbb{E}[\cdot] \leq n$.	Explain where $\sqrt{n}$ comes from
PART II	—	Not examined	—

EXP3 Proof Skeleton:

1. **Invoke Hedge:** $\sum_t \mathbf{v}_t \cdot \ell_t^h - \sum_t \mathbf{u} \cdot \ell_t^h \leq \frac{\ln n}{\eta} + \frac{\eta}{2} \sum_t \sum_i v_{t,i} (\ell_{t,i}^h)^2$.
2. **Take $\mathbb{E}$, three identities:**
 - $\mathbb{E}[\mathbf{v}_t \cdot \ell_t^h] = \mathbb{E}[\ell_{t,\hat{y}_t}]$ (algorithm's loss).
 - $\mathbb{E}[\mathbf{e}_i \cdot \ell_t^h] = \ell_{t,i}$ (comparator is deterministic).
 - $\mathbb{E}[\sum_i v_{t,i} (\ell_{t,i}^h)^2] = \sum_i \ell_{t,i}^2 \leq n$.
3. **Result:** $\mathbb{E}[L_A] - \min_i L_i \leq \frac{\ln n}{\eta} + \frac{\eta mn}{2}$.
4. **Optimize:** $\eta^* = \sqrt{2 \ln n / (mn)} \Rightarrow \sqrt{2mn \ln n}$. □

Key adaptation skill: If you observe k arms per round, second moment $\leq n/k$, so regret $\rightarrow \sqrt{2mn \ln n / k}$.

9 Cross-Cutting: Unified Proof Techniques

	Hedge	OGD	EXP3	Perceptron
Potential	KL $d(\mathbf{u}, \mathbf{v}_t)$	$\ \mathbf{w}_t - \mathbf{u}\ ^2$	KL (via Hedge)	$\ \mathbf{w}_t\ ^2$
Key trick	$e^{-x} \leq 1 - x + \frac{x^2}{2}$	Expand square	Take $\mathbb{E}$	Cauchy–Schwarz
Telescope	$\Phi_1 - \Phi_{m+1}$	$\ \mathbf{w}_1 - \mathbf{u}\ ^2$	(via Hedge)	—
Optimize η	$\sqrt{2 \ln n / m}$	$U / (R\sqrt{m})$	$\sqrt{2 \ln n / (mn)}$	N/A
Regret	$\sqrt{2m \ln n}$	$UR\sqrt{m}$	$\sqrt{2mn \ln n}$	R^2 / γ^2

Technique	Where Used	Core Idea
Union bound	PAC (realizable + agnostic)	Handle “ $\exists h$ ” by summing over all h
$(1 - x) \leq e^{-x}$	PAC realizable; AdaBoost; WM	Convert product to exponential
Hoeffding	Validation bound; agnostic PAC	Concentration of empirical average
Potential function	Hedge, OGD, WM, EXP3	Measure “distance to target”; telescope
Importance weighting	EXP3	Partial $\rightarrow$ full info via unbiased estimator
KKT conditions	SVM (hard + soft margin)	Characterize support vectors
Representer theorem	Ridge dual, kernel methods	Optimal $\mathbf{w} \in \text{span}\{\mathbf{x}_i\}$

10 Not Examined (Save Time)

- NFL proof (statement only)
- Boosting generalisation guarantees
- Kirchhoff Circuit Laws
- NP-hardness of sparse classifier
- VC dimension bound proof intuitions (Fundamental Theorem proof)
- Graph SSL Sections VIII–X
- EXP3 Part II

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